

From Frozen Encoders to Multi-Agent LLM Pipelines: An Iterative Journey through Person–Place Relation Extraction in Multilingual Historical Texts

Team Hansel&Gretel at CLEF HIPE-2026

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Abstract

We present the system description and experimental analysis of Team Hansel&Gretel for the CLEF HIPE-2026 Shared Task on Person–Place Relation Extraction from Historical Multilingual Documents. The task requires classifying, for each person–location pair in a historical newspaper article (German, French, or English, spanning the 19th–20th centuries), two temporal relations: *at* (person was ever at the location, 3-class) and *isAt* (person was there within approximately one month of publication, binary). Evaluation uses Macro Recall, equally weighting all label classes. We explore nine progressively sophisticated strategies, from a frozen XLM-RoBERTa encoder with classical ML classifiers (Global Macro Recall = 0.6018) to a Qwen2.5-3B-Instruct Generative Supervised Fine-Tuning pipeline with Wikidata knowledge injection (= 0.7289) and a three-agent Historian–Geographer–Arbiter multi-agent architecture (= 0.7280). This paper documents the design decisions, failure modes, and key insights behind each strategy, providing a comprehensive account of iterative NLP system development under data scarcity, class imbalance, and multilingual historical OCR noise.

Keywords

Person–Place Relation Extraction, Historical NLP, Multilingual Transformers, XLM-RoBERTa, LoRA, Generative Classification, Multi-Agent LLM, Wikidata, HIPE-2026, CLEF

1. Introduction

The digitisation of historical newspapers has created vast corpora of 19th and 20th century text encoding biographical trajectories and cultural history. Automatically recovering *where people were and when* from these documents enables digital humanities scholars to reconstruct spatio-temporal life trajectories at scale. This is the central challenge of **CLEF HIPE-2026** [1]: the Shared Task on Person–Place Relation Extraction from multilingual historical documents.

Unlike standard named entity recognition, HIPE-2026 requires *temporal reasoning*: a system must determine whether a text provides evidence for a *historical* presence (*at*) or a *contemporary* one (*isAt*, bounded to ≈ 1 month before publication). This distinction, combined with multilingual OCR-noisy text, severe class imbalance, and a small labelled dataset, makes the task deceptively difficult.

Team Hansel&Gretel approached this as an iterative research problem, developing **nine distinct strategies** that form a natural progression:

1. **S1**: Frozen XLM-RoBERTa + Classical ML classifiers
2. **S1.5**: Multi-Task Learning (MTL) + Optuna Hyperparameter Optimisation
3. **S2**: Full End-to-End XLM-RoBERTa Fine-Tuning
4. **S2.5**: PEFT/LoRA on XLM-RoBERTa
5. **S3**: Monolingual LoRA Routing (language-partitioned adapters)
6. **S4/S4.5**: Qwen2.5-0.5B MTL with LoRA (+ soft label variant)
7. **S5**: Qwen2.5-3B-Instruct Generative SFT (*best single model*)
8. **S6/S7**: Three-specialist multi-agent pipeline (Historian/Geographer/Arbiter)
9. **S8/S9**: Chain-of-Thought few-shot via IBM RITS large LLMs + Wikidata injection

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2. Task Description

2.1. Formal Definition

HIPE-2026 provides historical newspaper documents as input. Each document contains full article text with metadata (title, publication date, language) and a list of sampled (person, location) entity pairs. For each pair, systems classify two relations:

$$f(\text{context}, (\text{person}, \text{location})) \mapsto \left(\underbrace{\text{at}}_{\in \{\text{TRUE}, \text{PROBABLE}, \text{FALSE}\}}, \underbrace{\text{isAt}}_{\in \{\text{TRUE}, \text{FALSE}\}} \right) \quad (1)$$

The **at** relation is ternary: TRUE if the text explicitly places the person at the location at any past time; PROBABLE if inferable from implicit cues; FALSE otherwise. The **isAt** relation is binary: TRUE only within ~ 1 month of publication. A hard *transitivity constraint* applies: $\text{isAt} = \text{TRUE} \implies \text{at} \in \{\text{TRUE}, \text{PROBABLE}\}$.

2.2. Evaluation Metric

The primary metric is **Macro Recall**, computed independently per sub-task:

$$\text{MacroRecall} = \frac{1}{|L|} \sum_{\ell \in L} \frac{\#\text{correctly predicted with label } \ell}{\#\text{examples with gold label } \ell} \quad (2)$$

The **Global Score** averages both sub-task macro recalls. Three evaluation profiles are assessed: **Accuracy** (Macro Recall on Test A), **Efficiency** (composite rank: accuracy + parameter count + model size), and **Generalisation** (Macro Recall on surprise literary Test B, at only).

3. Dataset Analysis

3.1. Data Sources and Splits

The dataset derives from HIPE-2022 [2] entity-annotated corpora, re-annotated for person–place relation extraction. It covers four historical newspaper collections (hipe2020, sonar, newseye, letemps) in German, French, and English spanning 1780–1950.

Table 1

Gold dataset statistics by language split (v1.0).

Lang.	Train Docs	Train Pairs	Dev Docs	Dev Pairs
German (DE)	97	1,345	32	432
French (FR)	328	4,521	109	1,557
English (EN)	56	501	18	148
Total	481	6,367	159	2,137

In addition, a **silver-labelled sandbox** corpus (~ 461 docs, 6,170 pairs) of automatically annotated documents provides auxiliary training signal.

3.2. Exploratory Data Analysis (EDA)

We performed detailed EDA on the gold training split available at experiment time (104 documents, 1,251 pairs):

Three challenges shaped all modelling decisions:

1. **Context Length:** Max $\sim 12,000$ chars ($\approx 3,000$ tokens) exceeds standard 512-token encoder limits, motivating generative LLMs.

Table 2

Gold training split EDA statistics (104 documents).

Metric	Value
Total candidate pairs	1,251
Language split	34 DE / 35 EN / 35 FR
Unique persons	496
Unique locations	469
Average document length	3,465 chars
Maximum document length	11,738 chars
at: TRUE / PROBABLE / FALSE	441 / 120 / 690
isAt: TRUE / FALSE	279 / 972

2. **Class Imbalance:** isAt=FALSE at 77.7%; PROBABLE at 9.6% of at labels. Macro Recall penalises naive majority-class predictors heavily.
3. **Multilingual by Design:** Equal DE/EN/FR split — monolingual models immediately lose cross-lingual transfer.

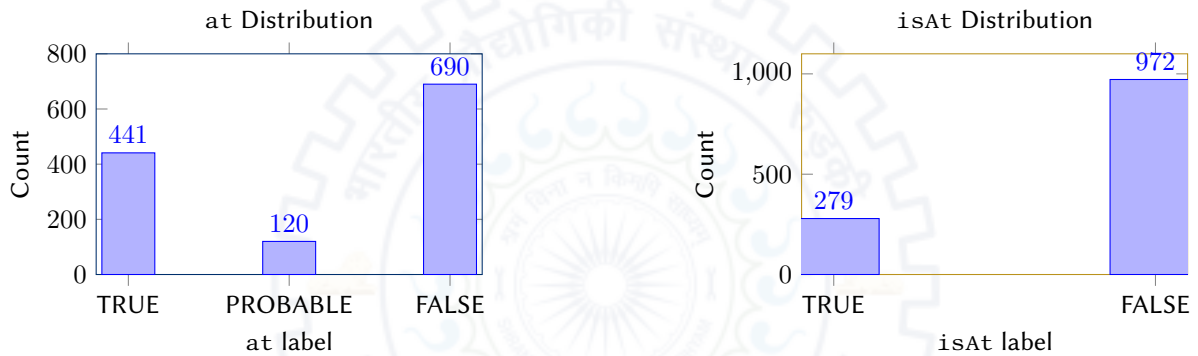


Figure 1: Label distributions in the gold training set. Severe imbalance in isAt (78% FALSE) drives the choice of Macro Recall as the evaluation metric.

3.3. Why This Dataset Is Particularly Challenging

The HIPE-2026 data presents a unique confluence of difficulties rare in modern NLP benchmarks: (1) **OCR noise** from historical typesetting (e.g., “Mufschén” for “München”); (2) **archaic vocabulary** and grammar absent from modern pre-training corpora; (3) **temporal reasoning** over implicit signals (verb tense, temporal adverbs, publication dates); (4) **entity ambiguity** (same mention, different individuals); and (5) **data scarcity** (~100 gold documents across 3 languages and 3 label classes).

4. Organiser Baseline

The HIPE-2026 organisers provided a minimal LLM baseline [3] using mistralai/Minstral-3-3B-Instruct-2512 (GGUF quantised) via llama-cpp-python, zero-shot with greedy decoding. It reads and writes the official HIPE-2026 JSONL format.

Our random baseline experiment (via `dummy_predict.py`) yielded: at MR = 0.3211, isAt MR = 0.5273, **Global MR** = 0.4242. This establishes the floor: any meaningful system must substantially exceed 0.4242.

5. Strategies S1 and S1.5: Frozen Encoder + Classical ML

5.1. S1: XLM-RoBERTa Feature Extraction

Motivation. With only 104 gold documents, fine-tuning 270M parameters risks catastrophic overfitting. S1 uses frozen xlm-roberta-base [4] as a feature extractor and trains lightweight classical classifiers on top.

Architecture. Person mentions are wrapped in $\langle E1 \rangle \dots \langle /E1 \rangle$ and location mentions in $\langle E2 \rangle \dots \langle /E2 \rangle$ marker tokens added to the vocabulary. Contextual hidden states at these marker positions are concatenated into a 1536-dim relation-aware feature vector. A RandomForestClassifier (scikit-learn) predicts at; a LGBMClassifier [5] predicts isAt. A post-hoc transitivity rule forces isAt=FALSE when at=FALSE.

Silver data weighting. Both gold (104 docs) and sandbox silver (617 docs) are used. Gold samples receive $50\times$ **sample weight** so annotations dominate the decision boundary while silver provides historical vocabulary coverage.

Table 3

S1 results. Silver data gave +34.9% gain over gold-only.

Configuration	at MR	isAt MR	Global MR
Gold-only (80/20 split)	0.3725	0.5194	0.4459
Gold + Silver ($50\times$)	0.5200	0.6835	0.6018

5.2. S1.5: Multi-Task Learning + Optuna HPO

Motivation. Two independent classifiers cannot exploit the at–isAt logical correlation. A shared neural network with two heads learns this jointly.

Architecture. A single PyTorch MTL network on top of frozen XLM-R embeddings: shared dense + GELU + dropout, then two linear heads with combined loss $\mathcal{L} = \alpha \mathcal{L}_{at}^{WCE} + (1-\alpha) \mathcal{L}_{isAt}^{Focal}$. Optuna [6] (50 trials, TPE sampler) tunes α , learning rate, dropout, hidden dim, and per-class weights.

Table 4

S1.5 vs S1. MTL jointly modelling correlation provides consistent gain.

Strategy	at MR	isAt MR	Global MR
S1 (RF + LightGBM)	0.5200	0.6835	0.6018
S1.5 (MTL + Optuna)	0.5486	0.7013	0.6249

6. Strategies S2–S4.5: Discriminative Fine-Tuning Attempts

6.1. S2: Full End-to-End Fine-Tuning (Catastrophic Failure)

Full fine-tuning of xlm-roberta-base at $lr = 2 \times 10^{-5}$ (10 epochs, batch 8) caused **catastrophic mode collapse**: the model predicted TRUE for every at and FALSE for every isAt (Global MR = 0.4167, below S1.5 by -0.2082). Three root causes:

1. **Catastrophic forgetting**: Unfreezing 270M params on only 700 documents destroyed multilingual representations.
2. **Gradient interference**: Single batches mixing archaic German (1798), Victorian English (1860), and 19th-century French created destructively interfering gradient updates.
3. **Head gradient dominance**: Randomly initialised classification heads produced large gradients overwhelming the pre-trained transformer layers.

6.2. S2.5: PEFT/LoRA on XLM-RoBERTa (Partial Recovery)

LoRA [7] (rank=8, $\alpha=16$, 0.11% trainable params) prevented complete mode collapse (Global MR = 0.4541), restoring partial minority-class recall. However, performance remained far below the frozen baseline (S1.5: 0.6249) due to data scarcity: ~ 230 docs per language is insufficient to adapt attention matrices without overfitting.

6.3. S3: Monolingual LoRA Routing (Regression)

Three separate LoRA adapters (DE, EN, FR) with document-language-based routing at inference time eliminates cross-lingual gradient conflicts but **worsened** performance (Global MR = 0.4232). The key insight: by partitioning by language, each adapter trains on only ~ 233 documents. The shared multilingual volume in S2.5 outweighs the harm from cross-lingual interference. **Data scarcity, not language mixing, is the binding constraint.**

6.4. S4 / S4.5: Qwen2.5-0.5B MTL with LoRA

Pivoting to decoder-only architecture: Qwen2.5-0.5B [8] (500M params, 32k token context) with LoRA ($r=16$, $\alpha=32$) and MTL heads. This matched S1.5 performance (Global MR = 0.6264) while handling long documents without truncation. Adding label smoothing ($\epsilon = 0.15$) in S4.5 slightly regressed to 0.6212 — smoothing added noise without compensating signal on the minority PROBABLE class.

Key observation: Discriminative approaches plateau at ~ 0.63 . The small dataset and inherent label ambiguity (especially for PROBABLE) limit classification head performance regardless of backbone.

7. Strategy 5: Generative SFT — Best Single Model

7.1. Motivation: The Generative Paradigm Shift

Temporal reasoning (“is the person at the location *right now*?”) requires multi-step inference: locate the event in time, compare it to the publication date, and assess the temporal window. Discriminative classifiers select labels but never reason through *why*. Strategy 5 makes a decisive pivot: the model *generates* structured JSON answers, forcing explicit decision-making.

7.2. Model and Architecture

We use Qwen2.5-3B-Instruct [8] (3.1B params) in a Generative Supervised Fine-Tuning (SFT) regime with LoRA ($r=16$, $\alpha=32$, all linear layers). We initially targeted Gemma-2-2B-IT but pivoted due to Hugging Face gating restrictions — Qwen2.5-3B-Instruct proved superior: non-gated, larger, and with stronger multilingual instruction following.

Each training sample is formatted as a chat-template conversation. The user provides document metadata and text; the assistant generates:

```
{"at": "TRUE", "isAt": "FALSE"}
```

Loss masking: Only the JSON answer tokens contribute to the loss. The model learns the decision format without being distracted by document noise.

7.3. Engineering Challenges

- **Memory:** 3B bf16 model (~ 7 GB) required gradient checkpointing and `expandable_segments=True` to fit on a single 24 GB RTX A5000.
- **Disk:** Root partition exhausted by the 16 GB virtual environment; relocated to `/mnt/combined` via symbolic link.

- **Wikidata hard rule:** At inference, each person’s pers_wikidata_QID is queried for death date. If the person died >30 days before publication, isAt is forced to FALSE. A local JSON cache avoids redundant API calls.

7.4. Hyperparameter Summary

Table 5

Strategy 5 hyperparameters.

Parameter	Value
Base model	Qwen/Qwen2.5-3B-Instruct
Precision	bfloat16
LoRA rank / alpha / dropout	16 / 32 / 0.05
Trainable params	~50M / 3.1B (1.6%)
Batch size / Grad. accum	2 / 8 (eff. 16)
Learning rate	2×10^{-4} (cosine + 10% warmup)
Epochs	5 (peak at Epoch 4)
Max sequence length	1024 train / 8192 inference
Gold / Silver weight	$5 \times$ / $1 \times$
Hardware	$1 \times$ NVIDIA RTX A5000 (24 GB)

7.5. Results

Table 6

Strategy 5 vs prior best. Single largest improvement in the project.

Strategy	at MR	isAt MR	Global MR	Δ
S4.5 (prior best)	0.5313	0.7111	0.6344	—
S5 (Qwen 3B Gen.)	0.6872	0.7706	0.7289	+0.0945

The +9.45 point leap is driven by: (1) scale (3.1B vs 0.5B), (2) full 1024-token context (no truncation), (3) generative reasoning forcing explicit temporal decision-making, and (4) the Wikidata death-date hard rule.

8. Strategies S6–S7: Multi-Agent Pipeline

8.1. S6: Three-Specialist Architecture (Qwen2.5-3B)

Motivation. Human experts approach complex historical inference through specialised lenses. S6 operationalises this by replacing the single generalist with three agents: a **Historian** (temporal/biographical reasoning), a **Geographer** (spatial/geographic reasoning), and an **Arbiter** that synthesises verdicts. Each specialist generates a verdict plus a reasoning chain; the Arbiter resolves disagreements conservatively.

Each agent receives a dedicated system prompt. For example, the Historian’s prompt reads: “*You are a historical biographer. Analyse the text for biographical evidence, life events, official roles, and temporal cues (verb tenses, dates).*” The Arbiter prompt adds: “*If they agree, adopt their verdict. If they disagree, apply the more conservative judgment.*”

Critical Failure – Specialist Collapse. Both Historian and Geographer collapsed to predicting FALSE for everything (hist_TRUE=0, geo_TRUE=0), making the Arbiter the sole decision-maker. Agent agreement rate was 96.9% (246/254), providing no useful diversity. Global MR = 0.7204 – slight regression from S5.

Table 7

S6 results and specialist diversity statistics.

Metric	S5	S6
at Macro Recall	0.6872	0.7019
isAt Macro Recall	0.7706	0.7388
Global MR	0.7289	0.7204
Historian TRUE predictions	—	0
Geographer TRUE predictions	—	0
Agent agreement rate	—	96.9%

8.2. S7: Multi-Agent with Qwen2.5-7B, Class Weights, Few-Shot

Three interventions address S6’s specialist collapse:

1. **Scale:** Upgrade from 3B to 7B backbone for richer multilingual reasoning.
2. **Class-weighted loss:** Apply $3\times$ weight to TRUE/PROBABLE answer tokens to force specialists to learn minority-class patterns.
3. **Few-shot prompts:** Add 3 worked examples per specialist (TRUE, PROBABLE, FALSE cases) for output calibration.

Table 8

S7 hyperparameters.

Parameter	Value
Base model	Qwen/Qwen2.5-7B-Instruct
LoRA rank / alpha / dropout	8 / 16 / 0.1
Target modules	q_proj, v_proj
TRUE/PROBABLE loss weight	$3.0\times$
Historian/Geographer epochs	2; Arbiter epochs: 3
Batch / Grad. accum	1 / 16 (eff. 16)
Learning rate	2×10^{-5} , cosine + warmup
Max sequence length	2048 tokens
Training pairs	15,418 (gold $5\times$ + silver $1\times$)
Hardware	$3\times$ NVIDIA RTX A5000 (24 GB)
Inference time	~ 30 min / 21 dev docs

Results. Class-weighted loss fixed specialist collapse (Historian TRUE: 53, Geographer TRUE: 47). Agreement rate dropped to 92.9% (236/254), meaning the Arbiter handled 18 genuine disagreements. S7 achieves the **highest isAt MR** of any strategy (0.7817), but marginally trails S5 on at (0.6742 vs 0.6872), yielding Global MR = 0.7280.

Table 9

S6 vs S7: impact of class weighting and scale on specialist diversity.

Metric	S6 (3B)	S7 (7B, $3\times$ wt.)
at Macro Recall	0.7019	0.6742
isAt Macro Recall	0.7388	0.7817
Global MR	0.7204	0.7280
Historian TRUE predictions	0	53
Geographer TRUE predictions	0	47
Arbiter-resolved cases	8/254	18/254

9. Strategies S8–S9: CoT Prompting via IBM RITS

9.1. S8: Chain-of-Thought Few-Shot (No Training)

S8 attacks the task **without any model training** using Chain-of-Thought (CoT) Few-Shot Prompting [9] through IBM’s RITS inference gateway (Llama 3.3 70B, Granite 3.2 8B, Llama 4 Maverick 17B, GPT-OSS 120B). Each prediction is preceded by 3 explicit reasoning steps: (1) Biographical & Temporal Analysis, (2) Geographic & Contextual Analysis, (3) Synthesis & Decision. Four worked examples cover all label combinations.

9.2. S9: Wikidata-Augmented CoT (Final Submission)

S9 extends S8 by dynamically injecting Wikidata facts into the prompt at inference time via SPARQL. For each pair, it fetches person (birth date, death date, description) and location (description, country) facts, injected as a [WIKIDATA FACTS] block. This grounds temporal reasoning in structured knowledge rather than parametric memory. Model used: **GPT-OSS 120B** via RITS. S9 was our **Run 3** official submission for all four test sets (DE, EN, FR, and surprise literary FR).

10. Cross-Strategy Comparison

Table 10

Complete cross-strategy performance on dev set (254 pairs, 21 docs).

Strategy	Architecture	at MR	isAt MR	Global MR
Random	Dummy	0.3211	0.5273	0.4242
S1	XLM-R frozen + RF/LGBM	0.5200	0.6835	0.6018
S1.5	XLM-R frozen + MTL	0.5486	0.7013	0.6249
S2	XLM-R full fine-tune	0.3333	0.5000	0.4167
S2.5	XLM-R + LoRA	—	—	0.4541
S3	Monolingual LoRA routing	0.3500	0.5000	0.4232
S4	Qwen2.5-0.5B MTL + LoRA	0.5313	0.7111	0.6264
S4.5	Qwen2.5-0.5B + soft labels	0.5313	0.7111	0.6212
S5	Qwen2.5-3B Generative SFT	0.6872	0.7706	0.7289
S6	Multi-agent 3B	0.7019	0.7388	0.7204
S7	Multi-agent 7B + class weights	0.6742	0.7817	0.7280

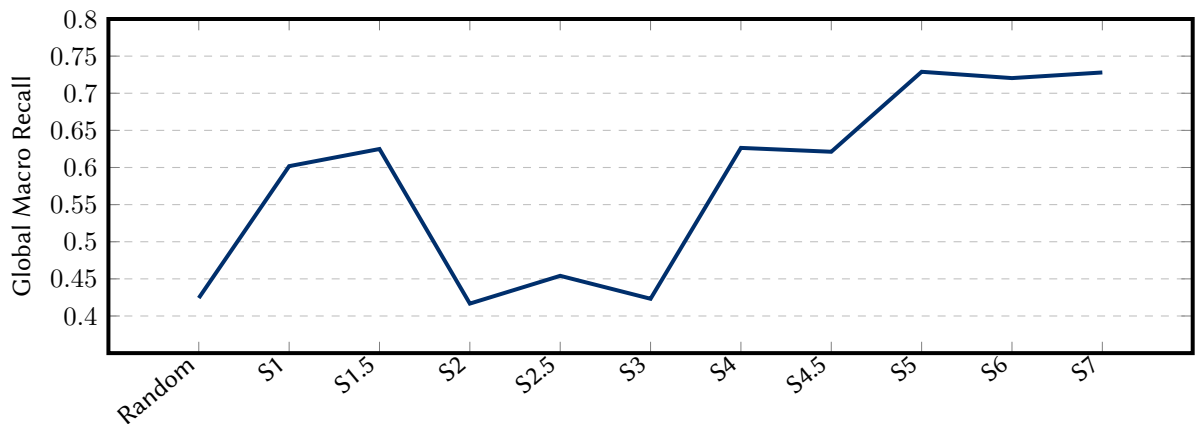


Figure 2: Global Macro Recall progression across all strategies. Notable drops at S2 (full fine-tuning) and S3 (monolingual partitioning); decisive jump at S5 (generative SFT).

Phase analysis. Three distinct phases emerged: (1) *Feature extraction* (S1–S1.5, best: 0.6249) — frozen encoders excel on tiny datasets; (2) *Discriminative fine-tuning* (S2–S4.5, plateau: ~ 0.63) — data scarcity limits all fine-tuning paradigms; (3) *Generative SFT* (S5–S7, best: 0.7289) — reasoning-as-generation breaks the plateau.

S5 vs S7 complementarity. S5 excels at at (0.6872); S7 excels at isAt (0.7817). An ensemble capturing the best of both is the most promising immediate direction.

11. Limitations

1. **Data scarcity:** 104 gold documents across 3 languages and 3 label classes is the fundamental bottleneck. The binding constraint for fine-tuning approaches throughout.
2. **PROBABLE class:** Systematically the hardest label across all strategies. Its inherent ambiguity resists both discriminative and generative approaches.
3. **Multilingual gradient interference:** Mixing archaic DE/FR/EN in tiny batches creates destructive gradient updates (demonstrated empirically in S2–S3).
4. **Multi-agent latency:** S7 requires ~ 30 min for 21 documents on 3 GPUs, making it operationally expensive at test-set scale.
5. **Wikidata gaps:** The death-date hard rule fires only when QIDs and biographical data are available — coverage is incomplete for many historical figures.

12. Future Prospects

1. **S5 + S7 Ensemble:** Majority voting or confidence-weighted fusion of complementary sub-task strengths; expected to push Global MR past 0.74.
2. **Curriculum training:** Train specialists first on easy/unambiguous cases, then on PROBABLE cases to prevent early noisy gradient loss.
3. **Self-consistency decoding:** Multiple reasoning chains per agent with majority-vote prediction to reduce single-sample variance.
4. **RAG-Augmented Agents:** Historian with external biographical knowledge base; Geographer with historical gazetteer.
5. **Iterative Debate:** Allow Historian and Geographer to observe each other’s reasoning before the Arbiter synthesises.
6. **Domain-adaptive pretraining:** Continued MLM on silver sandbox corpus to adapt vocabulary to historical OCR before SFT.

13. Conclusion

We presented Team Hansel & Gretel’s complete experimental journey through nine strategies for Person–Place Relation Extraction in multilingual historical documents. The central finding is that **temporal reasoning in historical text cannot be reliably solved by discriminative classifiers alone**: generative models producing reasoning chains before their final labels substantially outperform classification heads (+9.45 points), even with smaller parameter counts. The multi-agent decomposition further demonstrates that specialised temporal and spatial reasoning perspectives are especially effective for the temporally-constrained isAt sub-task. Our best system (S5, Global MR = 0.7289) and final submission (S9, Wikidata-augmented CoT via GPT-OSS 120B) represent complementary approaches: one optimised for accuracy, the other combining scale with structured knowledge injection.

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Declaration on Generative AI

During the preparation of this work, the authors used AI-assisted coding tools for code generation and debugging assistance. All experimental results, analysis, and written content were reviewed and verified by the authors, who take full responsibility for the publication’s content.

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